From LLMs to Multi-Agent Systems

A Progressive Guide to Agent Architectures

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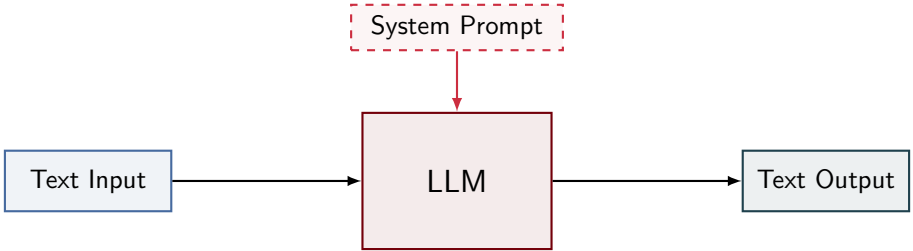
Outline

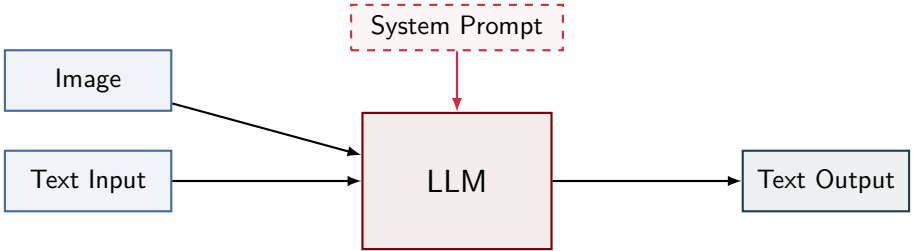
What Is an LLM?

- Text in, text out — a prediction engine
- No persistent memory between conversations
- System prompt shapes behavior (invisible to user)
- Size-capability tradeoff: small/fast vs. large/capable

LLM: Basic Interface



Multimodal LLM



Can see images, but output is still text only

AI-Assisted Coding (Autocomplete)

- LLM predicts code continuation
- No access to other files, tools, or execution
- User accepts or ignores suggestions

Before

```
def main():
    print("Hel|
```

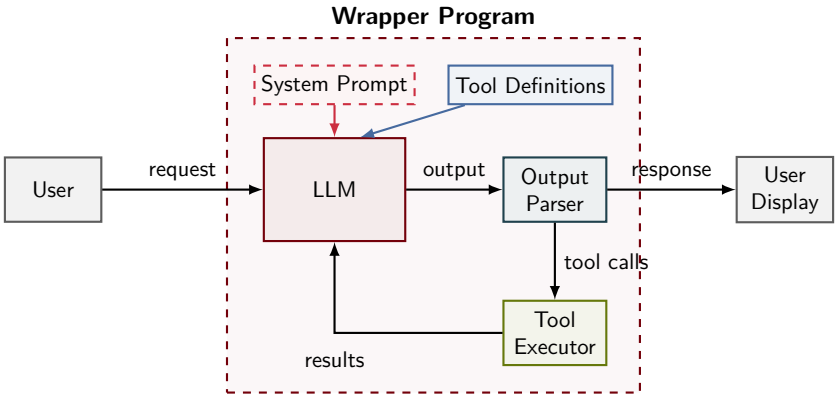
After

```
def main():
    print("Hello, World!")|
```

What Is an AI Coding Agent?

- LLM + wrapper program = agent
- Wrapper provides: tool definitions + execution environment
- LLM outputs tool calls → wrapper executes → results fed back
- Creates an action loop until task complete

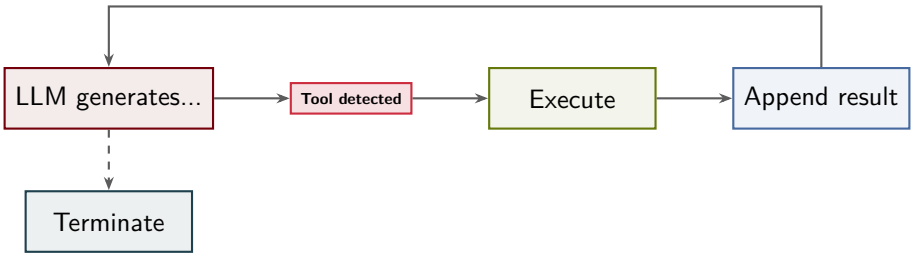
Agent Architecture



Three Execution Models

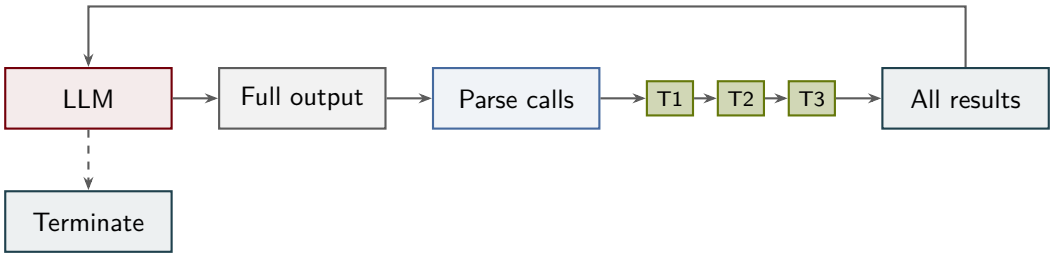
- **Streaming:** Execute tools as soon as detected mid-generation
- **Batch:** Generate full response, then execute all tools
- **Parallel:** Some tools run concurrently, others block

Streaming Execution



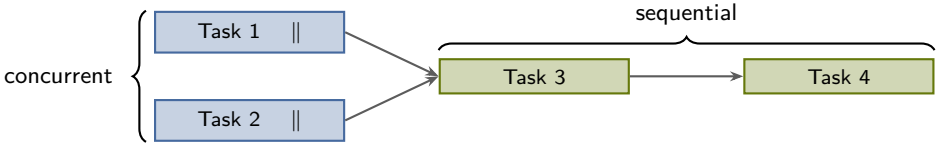
Real-time, responsive — tools fire immediately

Batch Execution



Simpler to implement — classic chatbot model

Parallel Execution

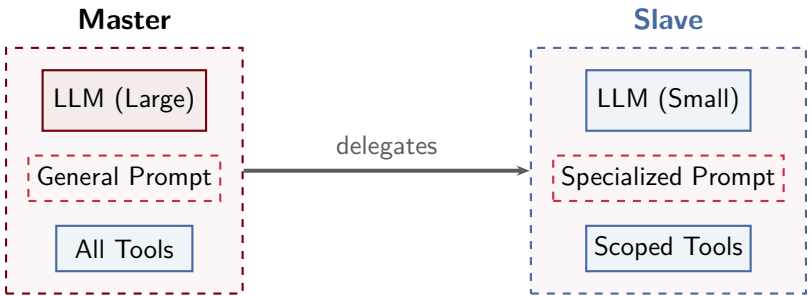


Independent tasks overlap; blocking tasks wait

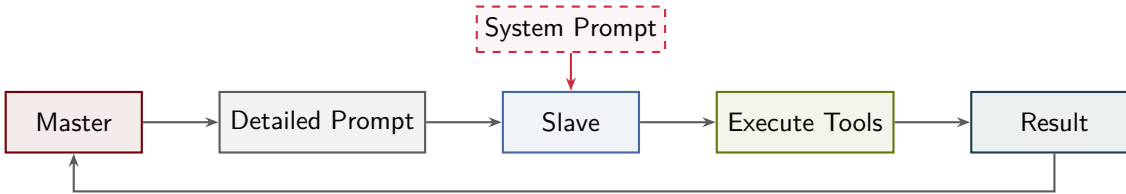
What Is a Slave?

- Same architecture as an agent
- Called by another agent (master), not by a human
- Often uses smaller/specialized model
- Scoped tools and specialized system prompt

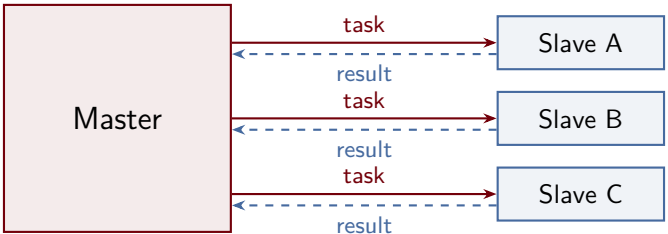
Master vs. Slave



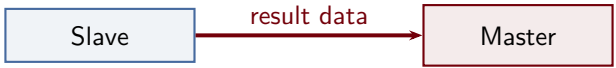
Slave Lifecycle



Master-Slave Pattern



Communication: Direct Return

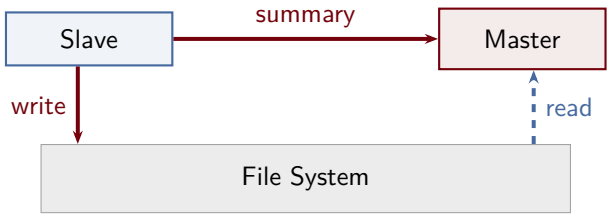


Result embedded in completion message
Best for small results (pass/fail, status codes)

Communication: File-Mediated

Best for large artifacts, binaries, generated files

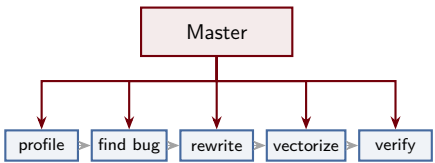
Communication: Hybrid (Most Common)



Detailed artifacts to disk + summary to master
Example: “14 passed, 2 failed” + full logs on disk

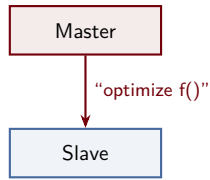
Imperative vs. Declarative Control

Imperative



many small slaves
one task each

Declarative



one capable slave
autonomous reasoning

Control Model Comparison

Property	**Imperative**	**Declarative**	**Hybrid**
Intelligence location	Master	Slave	Shared
Slave autonomy	Low	High	Medium
Predictability	High	Low	Medium
Flexibility	Low	High	Medium

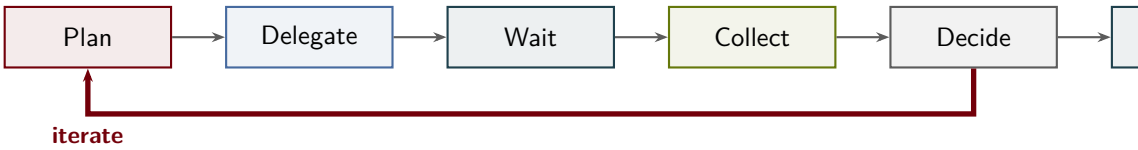
The Master Cycle

- Five-step loop: **Plan** → **Delegate** → **Wait** → **Collect** → **Decide**
- Differences lie in how each step executes
- May traverse once (single-pass) or repeat (iterative)

Single-Pass Orchestration

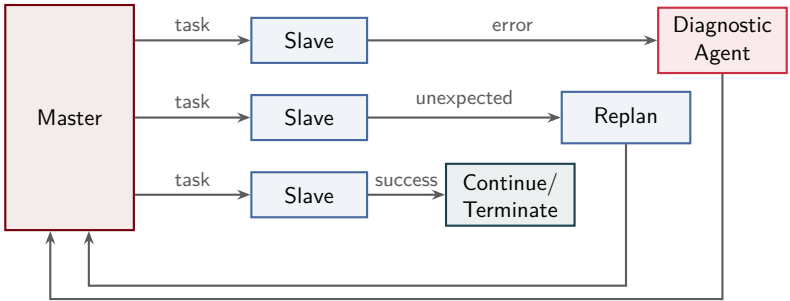
One traversal, no recovery, no refinement
If initial plan is wrong, no recourse

Iterative Orchestration



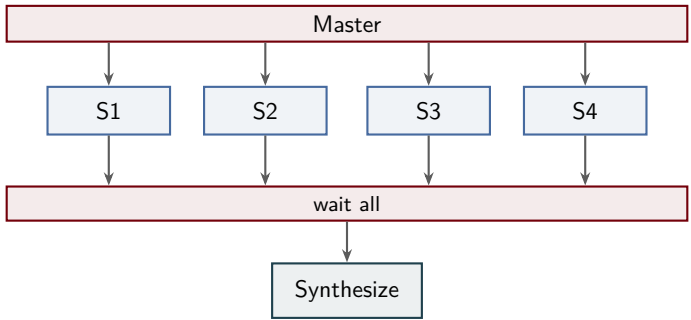
Feedback loop enables error recovery and refinement
Terminates on: quality threshold, max iterations, or no new tasks

Reactive Orchestration



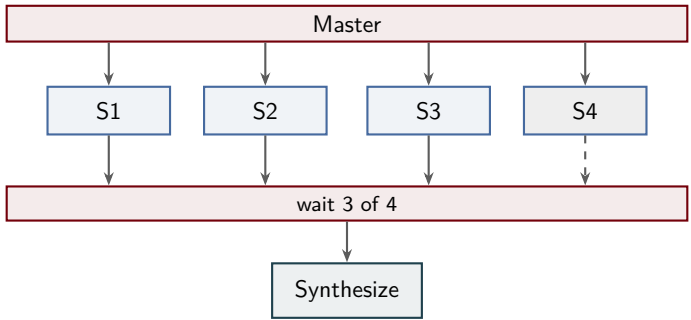
Event-driven: react to success, error, or unexpected discoveries

Parallel Agent Execution: Full Barrier



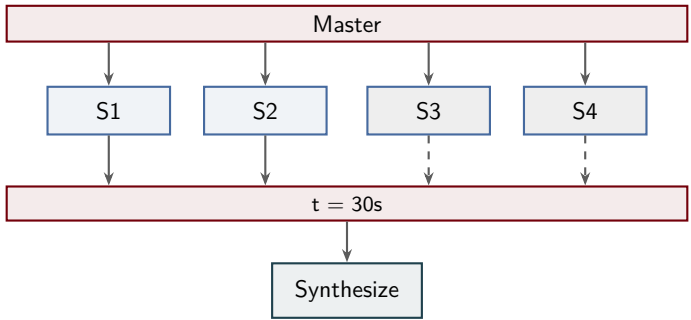
Slowest slave determines total time

Parallel Agent Execution: Partial Barrier (k-of-n)



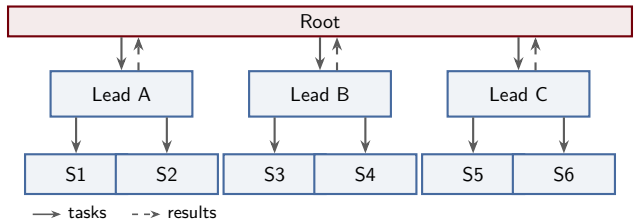
Proceed when k slaves finish; grayed slave not required

Parallel Agent Execution: Timeout Barrier



Prioritizes responsiveness over completeness

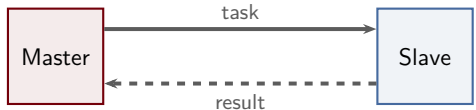
Tree-Structured Hierarchy



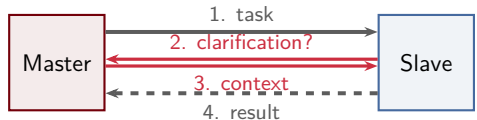
Tasks flow down, results flow up — root never talks to specialists directly

Unidirectional vs. Bidirectional Communication

Unidirectional:



Bidirectional:



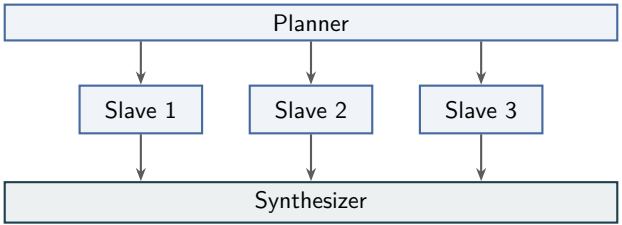
Four Canonical Patterns

- **Pipeline:** Serial, single-pass, unidirectional
- **Fan-Out/Fan-In:** Parallel, single-pass, unidirectional
- **Master-Slave Pool:** Parallel, iterative, unidirectional
- **Feedback Loop:** Serial, iterative, bidirectional

Pipeline Pattern

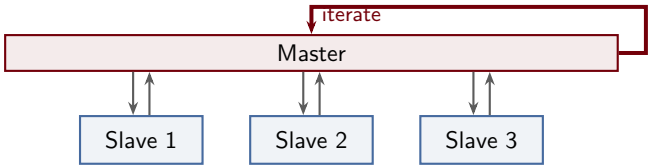
Sequential transformation chain
Latency = sum of all stages

Fan-Out/Fan-In Pattern



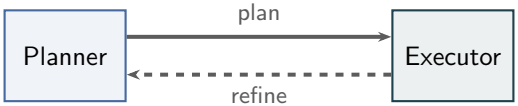
Independent subtasks in parallel
Latency = max of workers

Master-Slave Pool Pattern



Task queue + iterative processing
Best for batch workloads

Feedback Loop Pattern

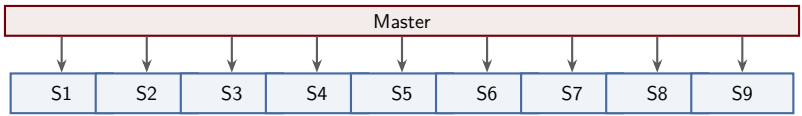


Iterative refinement until quality threshold
Best for tasks requiring multiple attempts

Why Multi-Layered?

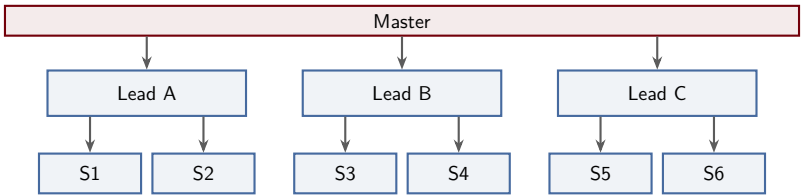
- Flat master = bottleneck at scale
- Context fills with operational details
- Layering separates: Strategy $\rightarrow$ Planning $\rightarrow$ Execution
- Each layer filters and abstracts information

Flat Architecture (Two-Layer)



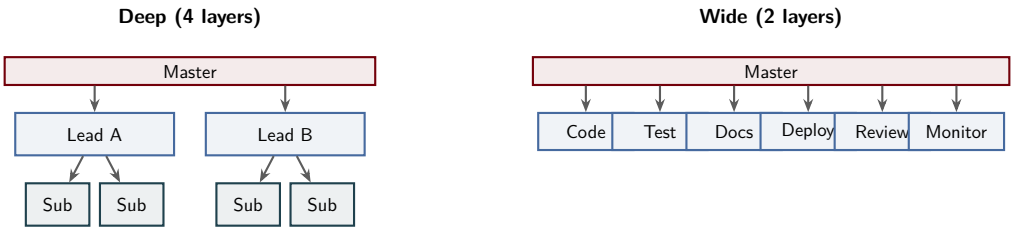
Master holds both strategic and tactical context
Context overload as slaves grow

Layered Architecture (Three-Layer)



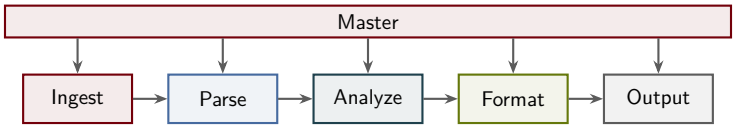
Each layer has smaller context budget
No single agent holds entire system state

Deep vs. Wide Architecture



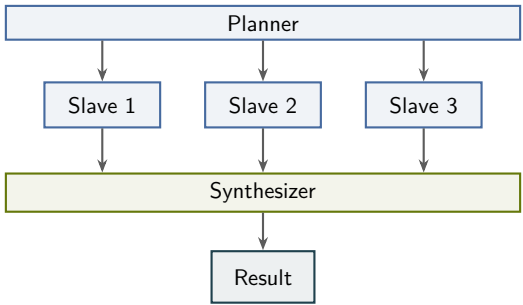
Deep: hierarchical refinement Wide: domain coverage

Pipeline Pattern in Practice



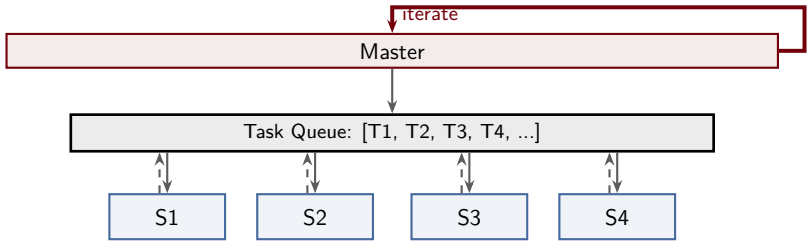
Document processing: each stage transforms and forwards

Fan-Out/Fan-In in Practice



Literature review: parallel searches, merged bibliography

Master-Slave Pool in Practice

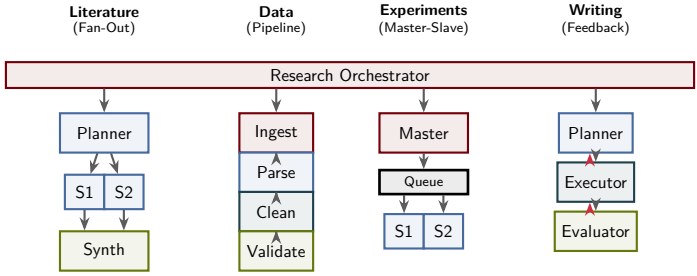


Batch processing: slaves pull tasks, results may generate new tasks

Feedback Loop in Practice

Code generation: write $\rightarrow$ test $\rightarrow$ fix $\rightarrow$ repeat

Composite Workflow Example

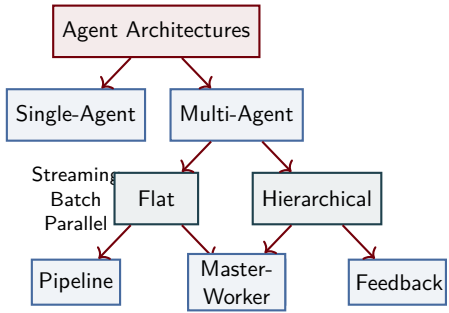


Real systems combine multiple patterns

Pattern Selection Guide

Criterion	**Pipeline**	**Fan-Out**	**Master-Slave**	**Feedback**
Independence	Low	High	High	Low
Iteration	No	No	Optional	Yes
Scale	3–5	3–20	10–100+	2–5
Latency	Sum	Max	Batched	Iterations
Best for	Transforms	Parallel work	Batch	Refinement

Agent Architecture Taxonomy



Key Takeaways

- LLM is an uncommitted tool — power comes from composition
- Four canonical patterns: Pipeline, Fan-Out, Master-Slave, Feedback Loop
- Architecture emerges from task structure, not model capability
- Patterns are composable — real systems combine them
- Multi-layered hierarchies scale beyond single-agent limits

Questions?

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